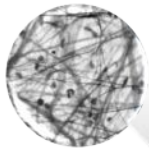
GENERAL BIOLOGY CONTENTS

Biology & Molecular Basis of Life.

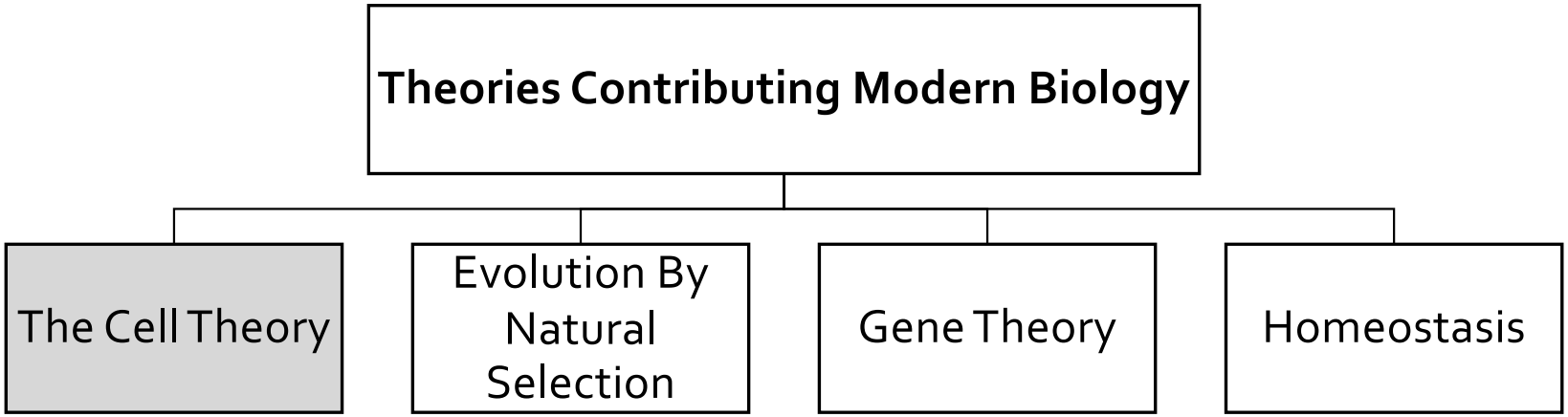
Cell Structure & Functions.

Introduction to Embryology.



Basic Histology.

Principles of Human Physiology.

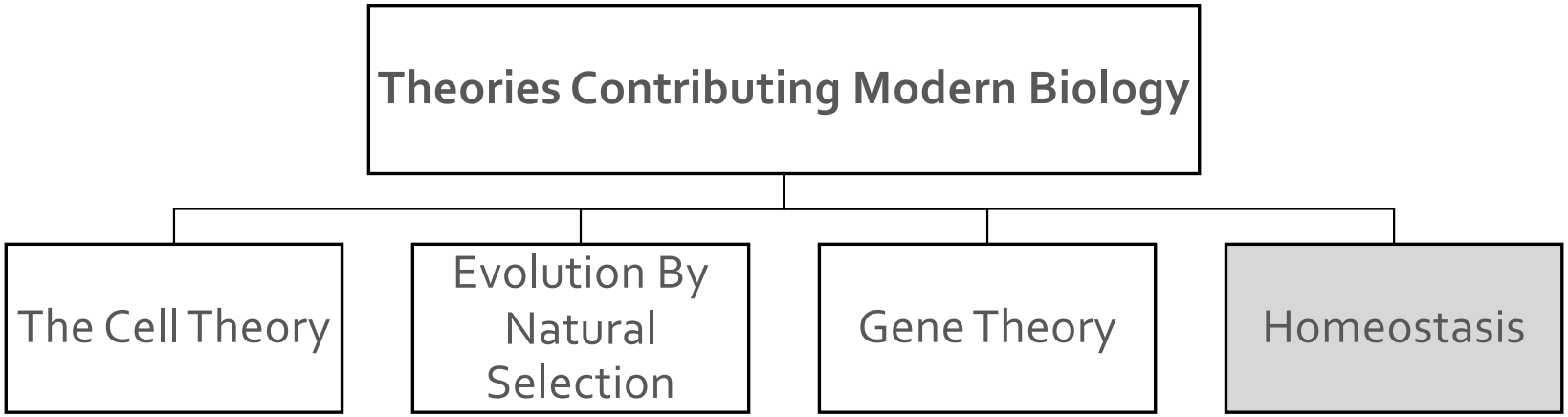


The Cell Theory

All living things are made of one or more cells.

Cells are the basic unit of structure and function in organisms.

Cells originate from existing cells.



Homeostasis

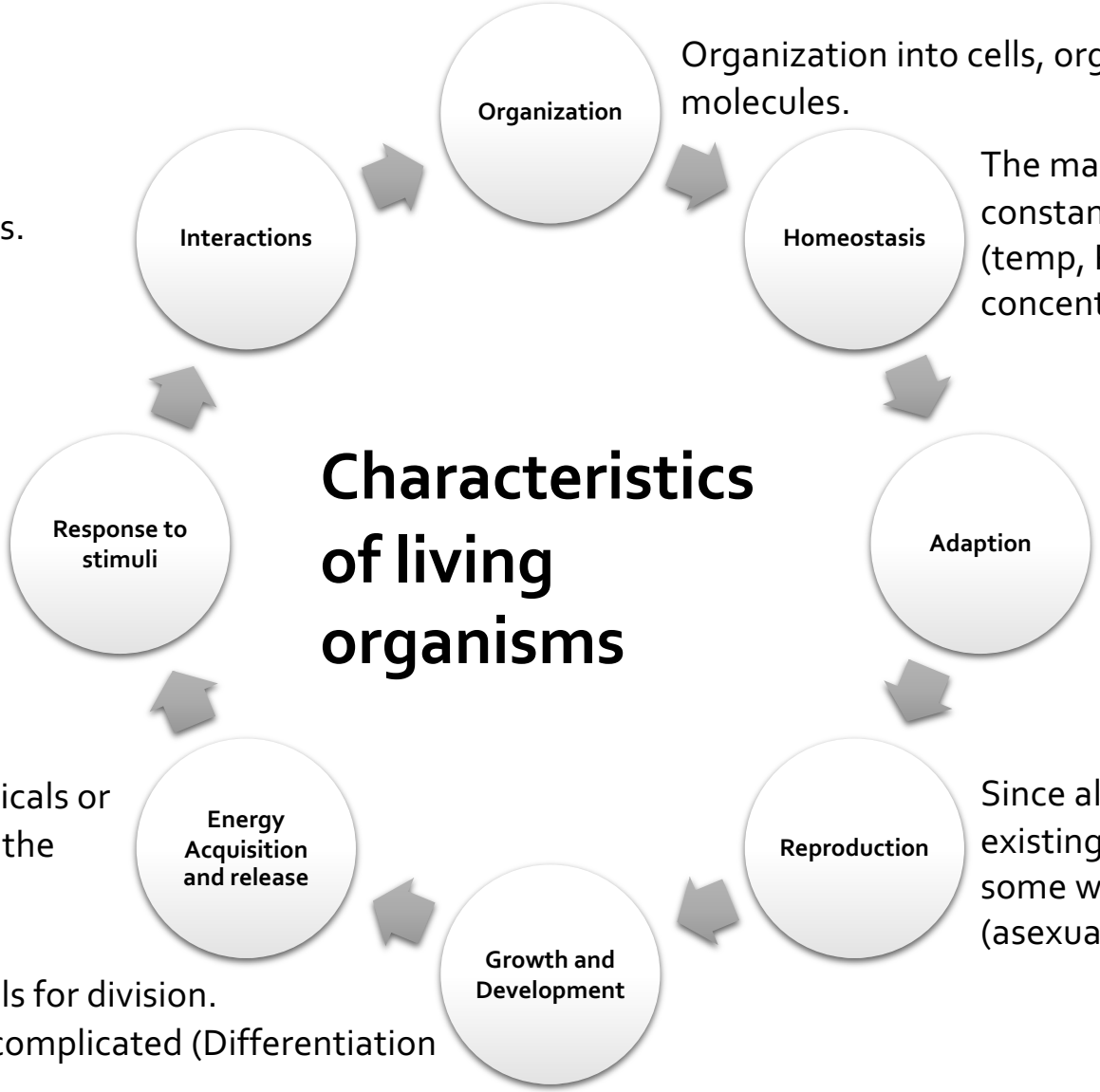
Homeostasis is the maintenance of a dynamic range of conditions within which the organism can function.

Temperature, PH and energy are major components of this concept.

Thermodynamics is the field of study that covers the laws governing energy transfers.

The universe is composed of two things: matter (atoms, molecules, organelles and cells) and energy.

Characteristics of living organisms



Organization into cells, organelles and molecules.

The maintenance of a constant internal environment (temp, PH, water concentration).

Living things are suited to their mode of existence.

Since all cells come from existing cells, they must have some ways to reproduce (asexual or sexual).

Single cells grow to mature cells for division. Multicellular organisms more complicated (Differentiation and Organogenesis).

From sunlight, inorganic chemicals or another organisms. Release in the form of ATP.

External or internal.

To Environment or each others.

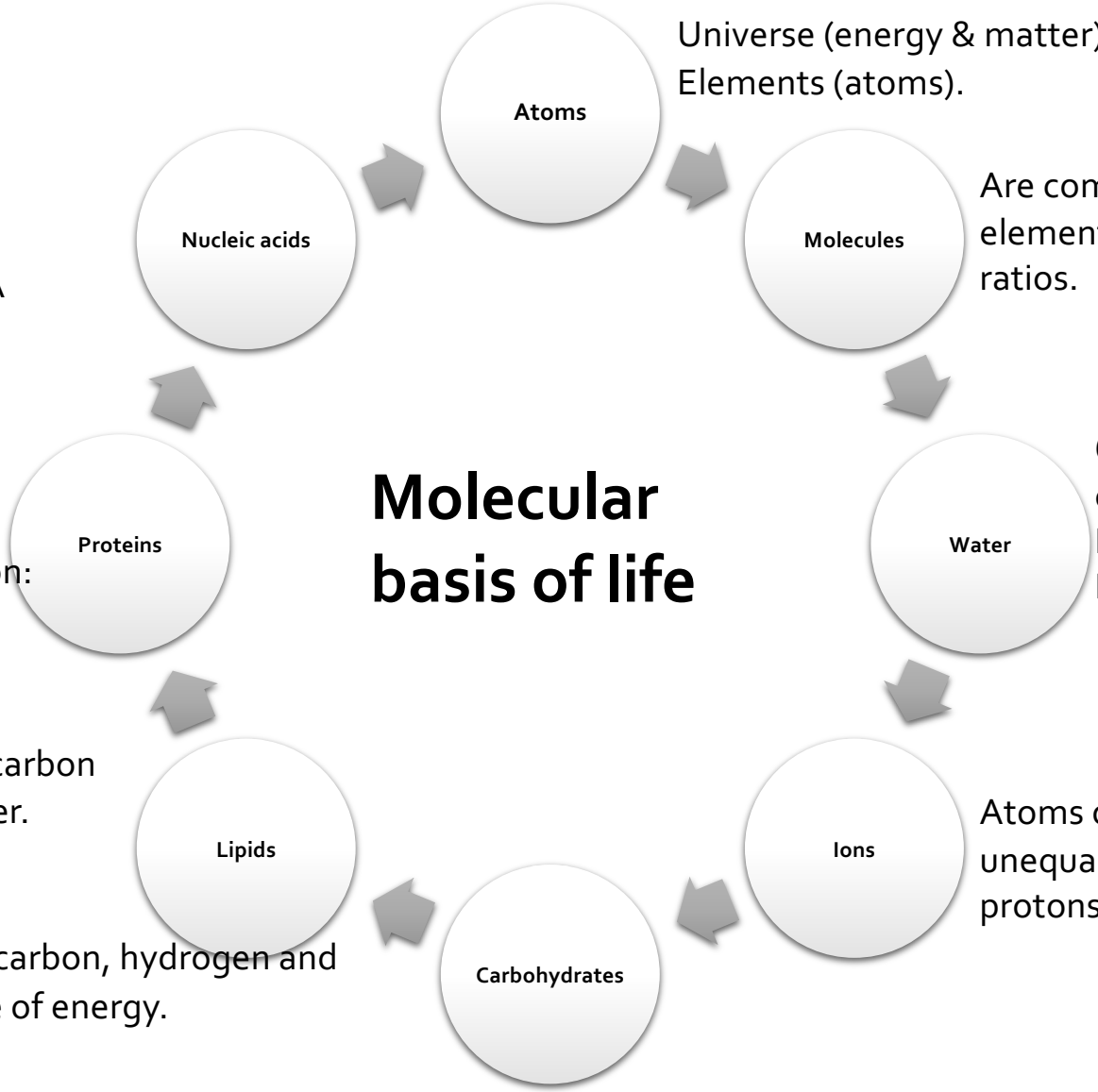
Levels of Organization

Biosphere	• The sum of all living things taken in conjunction with their environment.
Ecosystem	• The relation of a smaller group of organisms with each others and their environment.
Community	• The relation between groups of different species. Like dessert communities consists of rabbits, snakes and such dessert plants as cactus.
Species	• Group of similar individuals who tend to mate and produce viable, fertile offspring.
Population	• Group of similar individuals who tend to mate with each others in a limited geographic area.
Individuals	• One or more cells with a unique DNA arrangement. • These can be unicellular or multicellular.

Levels of Organization

Organ System	• In multicellular organisms. • A group of cells, tissues and organs performing a specific major function. Like cardiovascular system.
Organ	• In multicellular organisms. • A group of cells or tissues performing an overall function. Like heart.
Tissue	• In multicellular organisms. • A group of cells performing specific functions. Like heart muscle tissue.
Cell	• The fundamental unit of live things. • Each cells has DNA, energy acquiring chemicals and structures.
organelles	• A subunit of a cell, subcellular function. like ribosomes (protein synthesis), Mitochondria (ATP generation).
Molecules	• The fundamental functional levels of biochemistry.

Molecular basis of life



Universe (energy & matter), Matter (Elements), Elements (atoms).

Are compounds in which the elements are definite, fixed ratios.

60-90% living organisms. Excellent solvent & Metabolite reactions.

Atoms or molecules with unequal no. of electrons or protons.

3% dry mass of cell. Composed: carbon, hydrogen and oxygen atoms. Important source of energy.

40% of dry mass of cell. From: carbon and hydrogen. Insoluble in water. Structural component.

50-60% of dry mass of a cell. Composed: amino acids. Function: structure and enzymes.

DNA and RNA